

Image Contrast Enhancement using Fuzzy Logic with Metaheuristics

Mohimenul Kabir

Department of Computer Science
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
mahibuet045@gmail.com

Ahmad Hassanat

faculty of information technology
Mutah University
Mutah, Karak, Jordan
ahmad.hassanat@gmail.com

Jaiaid Mobin

Rochester Institute of Technology
Rochester, USA
jaiaidmobin@gmail.com

M Sohel Rahman

Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
sohel.kcl@gmail.com

Abstract

In this work, we present a method to enhance image contrast using a fuzzy logic-based transformation function. Our goal is to develop an image-specific transformation by tuning the fuzzy system. We employ Hill Climbing and Genetic Algorithms in various configurations to evolve the fuzzy system, conducting multiple experiments. Different variants of our method are tested on several images, and the two variants with the highest fitness scores are selected. Additionally, we compare against existing contrast enhancement technique and conduct a survey to evaluate the visual improvements made by these two variants. The survey results indicate that one method effectively enhances image contrast visually.

Keywords

Image Enhancement, Metaheuristics, Fuzzy Logic

ACM Reference Format:

Mohimenul Kabir, Jaiaid Mobin, Ahmad Hassanat, and M Sohel Rahman. 2024. Image Contrast Enhancement using Fuzzy Logic with Metaheuristics. In *11th International Conference on Networking, Systems, and Security (NSysS '24)*, December 19–21, 2024, Khulna, Bangladesh. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3704522.3704542>

1 Introduction

Image enhancement is the process of improving an image's quality and information content [45]. The goal is to increase visual differences among image features, making it more suitable for various applications, such as increasing the brightness of dark images for better viewing. Common image enhancement techniques include sharpening, smoothing, contrast enhancement, and noise reduction [4, 10, 33, 41]. Contrast enhancement, in particular, is applied to images or videos to increase their dynamic range [15].

Digital image processing is widely used across various applications, but uncontrollable factors during image acquisition can lead

to low-quality images [1, 51]. These low-quality images are often unsuitable for human observation and machine perception, making image enhancement a crucial task in computer vision, machine vision, and pattern recognition [19, 21, 54].

Metaheuristic techniques have been applied to several real-world computational problems across various domains [3, 20, 23–25]. Similarly, image enhancement has long been treated as a suitable problem for metaheuristics [17, 31, 32, 53]. Moreover, the integration of fuzzy systems with metaheuristic algorithms has been the subject of area within the Computational Intelligence community [8, 11]. Many studies have approached image enhancement as an optimization problem, focusing on modifying the image quality fitness function [10, 34, 53], combining several metaheuristic algorithms [17, 53], optimizing parameters [39] and avoiding local optima [9]. However, one aspect that has not received as much attention in image enhancement problems is the adaptation of fuzzy systems. Sandeep and Samrudh [22] have demonstrated that the variation in the input membership function in a fuzzy system can positively impact image enhancement performance.

This paper addresses the image contrast enhancement problem by using stochastic optimization within a fuzzy logic system. Our primary contribution is the design of a metaheuristic inspired framework that leverages a fuzzy logic system. The core component of the fuzzy logic system is a set of *input membership function* that are used to describe an *intensity transformation function*. We implement genetic operators that *mutate* the fuzzy logic system, thereby enhancing the original image to achieve optimal contrast enhancement. The fuzzy system in our approach is based on simple contrast enhancement rules, outlined in [13], and it operates independently of the input image. Our fuzzy image enhancement technique begins with a basic fuzzy rule set and a set of input membership functions. Using a metaheuristics framework, we evolve these fuzzy sets. The transformation function defined by the evolved fuzzy sets is then applied to the value channel of the HSV color space to produce the final enhanced image [40]. This approach converts the problem into an optimization challenge. Instead of directly generating an image, we aim to create a suitable mapping between input and output color values. This task is inherently difficult because, even with only 8-bit color (256 color values), the number of possible mappings is enormous. Exhaustive search through this vast solution space is infeasible, and there is no clear knowledge on how to improve or



This work is licensed under a Creative Commons Attribution International 4.0 License.

NSysS '24, December 19–21, 2024, Khulna, Bangladesh
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1158-9/24/12
<https://doi.org/10.1145/3704522.3704542>

generate a solution. Therefore, we leverage a metaheuristics framework to address this challenge [29]. Additionally, we empirically demonstrate the effectiveness of our proposed metaheuristic-based techniques in enhancing image contrast. Quantitative comparisons and a comprehensive survey reveal that the genetic algorithm technique, in particular, significantly improves the visual quality of images on par with existing image enhancement techniques.

The paper is structured as follows: Section 2 reviews the literature of image enhancement, Section 3 presents the background to understand the methodology of the paper, Section 4 formally presents our problem statement, Section 5 presents our methodology for enhancing image contrast, Section 6 empirically evaluates our proposed methodology for image enhancement, finally, we conclude our work with some indications of future works in Section 7.

2 Literature Review

Several techniques are used for contrast manipulation, including gamma transformation [18, 45] and histogram equalization (HE) [46, 47], but it has limitations, such as failing to maintain image brightness [42]. Bi-histogram Equalization [35] addresses this issue. Another drawback of HE is the potential loss of image information [55]. Techniques like gamma transform and log transform [45] offer lower computational complexity. Tarawneh et al. [49] applied gamma transformation for contrast enhancement by automatically applying different gamma corrections to multiple pixel sets. Huang et al [18] applied gamma correction with *weighting distribution*. However, these techniques often struggle in complex illumination settings [34], requiring parameter adjustments to be effective. Bhandari et al [6] combined sub-histograms with *discrete cosine transform* for contrast enhancement.

Fuzzy logic and metaheuristic techniques have been previously applied to image enhancement problems [8, 11, 26]. A fundamental approach involves using a fuzzy logic-based system from [13], which utilizes only three rules and employs trapezoidal and triangular input fuzzy sets. Besides, Joshi and Kumar [22] proposed a complex method with a seven-rule set and *gaussian* fuzzy sets.

To evaluate the fitness of an enhanced image, Munteanu and Rosa [32] proposed a novel objective function and applied an evolutionary algorithm to search for optimal parameters in a continuous transform function. This objective function has also been utilized in various optimization techniques, including artificial bee colony optimization [10], the cuckoo search algorithm [5], and the firefly algorithm for enhancing UAV-captured images [39].

Asamoah et al. [2] exploited some metrics to evaluate image enhancement techniques, including Mean Square Error (MSE), Peak Signal Noise Ratio (PSNR), Root Mean Square Error (RMSE), and Image Quality Index (IQI). PSNR metric is commonly used to assess the visual quality of images. This work also reported that out of multiple variants, histogram equalization (HE) has performed best in all 8 compared metrics. Inspired by this, we have also presented the PSNR metric comparison with baselines.

3 Preliminaries

Fuzzy Image Processing

Fuzzy image processing encompasses a range of techniques designed to understand, represent, and process images, along with their segments and features, as *fuzzy sets*. The process begins with

the *fuzzification* (or encoding) of image data, followed by the *defuzzification* (or decoding) of the results. These steps enable the application of fuzzy techniques to image processing. A *membership modification* [16] technique lies at the core of fuzzy image processing.

Fuzzy Sets for Intensity Transformation

Image contrast refers to the variation of intensity within an image [13, 14, 36]. Higher contrast implies greater variation. Contrast enhancement is a key application of intensity transformations, where increasing contrast can be achieved by making dark regions darker and bright regions brighter. This concept of enhancing contrast can be expressed through the following rules for a grayscale image [13]:

- IF a pixel is **dark**, THEN make it **darker**
- IF a pixel is **gray**, THEN make it **gray**
- IF a pixel is **bright**, THEN make it **brighter**

The concepts of **dark**, **gray**, and **bright** are inherently imprecise. To address this, we need to define what constitutes a **dark**, **gray**, or **bright** pixel. Each pixel is a member of a fuzzy set, and its degree of membership is determined by a membership function. Figure 1 illustrates the membership functions for the fuzzy sets as described in [13]. The dark and bright sets are clearly defined for very low and high-intensity values, respectively (z -axis). However, in the middle range of z , the definitions become less precise. The membership value for a specific z value is obtained from the y -axis, which ranges from 0 to 1, where 0 indicates “no membership” and 1 indicates “complete membership”.

Defuzzification refers to converting the fuzzy value to a *single crisp* value. For any given input pixel z_0 , the output crisp value v_0 will be as follows.

$$v_0 = \frac{\sum_{i=1}^n \mu_i(z_0) * v}{\sum_{i=1}^n \mu_i(z_0)}; n = 3 \quad (1)$$

$\mu_i(z_0)$ = fuzzy level of the image pixel with value z_0
 v = constant/mutable value used in the defuzzification

Color Information in Image

The color information of an image represents what color will be perceived by viewers. A commonly used color representation is RGB (Red-Green-Blue) where each pixel has three values. Each value indicates the intensity of three color channels: Red, Green, and Blue. Typically 8 bit is used for each channel, which means the value of [0–255] is used to represent the intensity. The overall color is the combination of the intensity of these three channels. This color model is suitable for color representation in display devices. As display devices generate each three colors independently and their combination gives us different color perceptions.

Another way to model color is HSV (Hue-Saturation-Value). HSV also uses three values for each pixel. The Hue channel is used to represent only the dominant wavelength (not its intensity or how pure it is). Saturation indicates how pure it is. For example, the pink color in the HSV model is a red hue but not pure red. It has a mix of other wavelengths hence the whiter look. A complete Saturation value means only the Hue-represented wavelength is giving the

perception [50]. The value channel indicates the intensity [44, 48]. A red hue can be perceived as black as if it has zero intensity. In OpenCV implementation of HSV color space, the range of Hue channel is $[0 - 179]$, Saturation and Value have range of $[0 - 255]$ [27]. Another thing to notice is that in HSV, we separate intensity information from color information [28].

4 Problem Statement

Our objective is to manipulate image contrast to enhance the sharpness and make image features more visually distinguishable. While image enhancement is inherently subjective, we have employed a fitness function from image enhancement literature to quantify this enhancement [5, 7, 9, 10, 32]. This fitness function evaluates the quality of an enhanced image. We use a transformation function to enhance the image, which is optimized using a metaheuristic approach. Thus, the input to our process is an image, and the output is an enhanced version of that image.

For simplicity, we present a formal definition of the problem in the context of a *grayscale* image. Let $I = f(x, y)$ represent a grayscale image, where x and y denote the pixel positions. The image I has dimensions $M \times N$, thus $0 \leq x < M$ and $0 \leq y < N$. Assume there is a fuzzy logic-based transformation function $T(I)$ that modifies the gray value of each pixel, producing an enhanced image I_e . More formally,

$$I_e = T(I) = T(f(x, y)) \quad (2)$$

Thus, our task is to identify a suitable transformation function that visually enhances the resulting image I_e .

5 Methodology

In this section, we present our metaheuristic-based techniques for image contrast enhancement. All metaheuristic techniques share several common features [12]. Each technique begins with an initial set of *individuals*, often referred to as the *population*. To introduce diversity into the population, special operations, known as *mutations*, are applied to generate new populations. In each iteration, the fittest individuals are selected based on some mathematical functions known as the *fitness function*. This process of generating a new population is repeated for a fixed number of iterations or a fixed time frame, with the fittest population from all iterations is considered the best solution found. While this approach does not guarantee an optimal solution, it is effective at discovering *good* solutions within a relatively short time frame. Finally, the differences between most metaheuristics techniques lie in the way in which the initial population is generated and the following populations are manipulated.

At a high level, we employed both Hill Climbing and Genetic Algorithm-based techniques with problem specific mutation and crossover operations. Now we discuss each of the subroutines of our proposed metaheuristic techniques.

5.1 Population Representation

Every metaheuristic technique begins with an initial population, usually randomly generated, and the representation of this population is specific to the problem being addressed. In our problem,

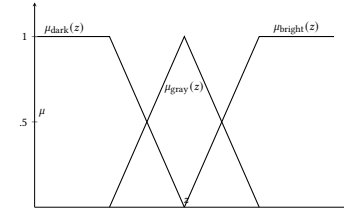


Figure 1: Input membership functions for fuzzy, rule-based contrast enhancement [13]

each individual represents an input membership function, as illustrated in Figure 1. The initial population thus contains information about a set of input membership functions. Each input membership function comprises three mathematical functions, each of them is represented by a tuple of three values, regardless of whether the function type is trapezoidal or triangular. The first two values of the tuple determine the shape of the function, while the third value is used in defuzzification.

5.2 Fitness Assessment

Following the literature of image enhancement [32], we calculate an individual's (I) quality/fitness using the following fitness function:

$$\text{Fitness}(I) = \log \log E(I_s) \times \frac{\text{ne}(I_s)}{M \times N} \times H(I_s) \quad (3)$$

I_s = image after Applying *Sobel filter* on I

$E(I)$ = sum of intensity of image I

$\text{ne}(I)$ = number of edge pixel in I

$H(I)$ = entropy of image I

M = width of I

N = height of I

5.3 Mutation

When generating a new individual, only the shapes of the functions (i.e., the width of the triangle and trapezoid's oblique line's slope) are tweaked. For implementation, we have adapted three hyperparameters, namely, ChangeProb, MutateMu, and MutateSigma. ChangeProb denotes a probability and during each iteration, a membership function is tweaked with a probability of ChangeProb. Whenever a membership function is tweaked, it is changed by some amount δ , and the value of δ is sampled from a gaussian distribution with mean and variance MutateMu and MutateSigma, respectively.

5.4 Crossover

The crossover operation combines multiple individuals (typically two) to produce offspring. There are three classical methods of performing crossover, and in our implementation, we have used *uniform crossover*. In the context of our representation, the crossover operation iterates through all the membership functions and combines them by swapping individual functions based on the outcome of a probabilistic coin toss with probability p . For the crossover in the Genetic Algorithm, both individuals must be of the same size.

5.5 Hill Climbing: Details

The Hill Climbing technique stochastically generates new individuals and stores the fittest individual as the candidate solution. The best solution is evaluated using the fitness function, as shown in Equation 3. In each generation, we generate a certain number of neighbor individuals (set to 10) from a fixed individual, following the mutation operations outlines in Section 5.3.

5.6 Genetic Algorithm: Details

Unlike Hill Climbing, the Genetic Algorithm (GA) starts with a number (say, PopSize) of individuals. We use the same evaluation function shown in Equation 3 for fitness evaluation in our GA approach. To introduce variations in the population, GA breeds a new population of children, selects individuals from the old population, and tweaks them to breed new individuals. To keep the footprints of both populations, it joins the parent and children populations to form a new generation of the population (Section 5.4). In our implementation, we have fixed PopSize equal to 30. The genetic algorithm differs in how parent and child populations are joined. In our genetic algorithm procedure, we have experimented with both (P, P) and $(P + P)$ evolution strategies, where P is the PopSize.

Hyperparameter Settings. We run each of the variants for 10 minutes and select the best solution found within that time frame. The parameters are set as follows: ChangeProb = 0.5, MutateMu = 3, and MutateSigma = 2.

6 Experimental Results

In this section, we evaluate our proposed approach and present the results of applying our methods to various images. Our experiments were conducted on several images from [10] and additional images (color, grayscale, and text) sourced from the Internet. Building on previous works in image enhancement [5, 10, 39] and to conduct a comprehensive evaluation, we experimented with a total of 18 images. For color images, we convert them to the HSV format and change the contrast in the image by manipulating brightness without affecting the color. However, we observed that our proposed method performs particularly well on grayscale images. Therefore, we present the results specifically for grayscale images (13 images). Our experiment study reveals that the genetic algorithm with (P, P) strategy outperforms the genetic algorithm with $(P + P)$ strategy. Thus, we present the results of genetic algorithm with (P, P) strategy. In the further analysis, we consider the technique:

- (1) (HC) Hill Climbing (Subsection 5.5)
- (2) (GA) Genetic Algorithm with (P, P) strategy (Subsection 5.6)

6.1 Experimental Setup and Environment

To evaluate the efficacy of our approach, we implemented a prototype in Python using the DEAP evolutionary computation framework¹. For the fuzzy logic component, we utilized the scikit-fuzzy toolbox². All variants of our algorithms were executed on a system with an Intel(R) Core(TM) i3 CPU M370 @ 2.40GHz processor, 6GB RAM, running Ubuntu 18.04, and Python version 3.6.8.

¹<https://deap.readthedocs.io/en/master/>

²<https://pythonhosted.org/scikit-fuzzy/>

6.2 Results

Due to the space limitation, we are unable to include all the enhanced or generated images in the paper. We first present the attained result by pointing out changes in features of four images. Figures 5f, 3f, 4f, 6f show output images generated by our technique against the reference input images. As an outcome of our technique, the enhanced images are more natural looking in the case of Figures 3f, 4f. In Figures 5f, 4f and 6f we can see the edges are sharper (notice the house shadow and roof tiles, cloud at top of jetplane and mountain body). Overall the histogram is expanded, and the contrast between the black and white portion has increased. Also, it is visually noticeable that method GA is doing better than HC. We will see support behind this claim in Section 6.3. A downside of our method is that some black portions (e.g., Figures 4e, 4f) are unnecessarily darkened. This can be beneficial in the case of overexposed images. For example, both methods in Figure 3 and 6 are succeeded in increasing the contrast between the person's background, face, and contrast between letters and page.

For quantitative comparisons, we compute the Michelson contrast [30, 37], PSNR [2], and SSIM [52] of each enhanced image in Table 1 through Hill Climbing (HC) and Genetic Algorithm (GA) variants of the proposed methods. We compare similar metrics with images enhanced through techniques: histogram equalization (HE), *dual channel* prior-based nighttime low illumination image enhancement (Dual) [43], and adaptive gamma correction with weighting distribution (AGCWD) [18].

The higher value of contrast indicates better contrast by increasing the difference between the darkest and brightest pixels. The table demonstrates that our proposed methods (especially GA) are effective in image contrast enhancement in most of the images compared to the original image. HE, Dual, and AGCWD have achieved better contrast metric value. Upon visual inspection, it is revealed that these three methods manage to increase contrast by darkening the dark region and brightening the bright region. In all presented images, Dual and AGCW affect oppositely compared to our method, that is they overexpose the already bright area in the images. This explains why HE, AGCW, and Dual is doing well in the Michelson contrast metric. Another observation is that out of all 5 methods, HE introduces noticeable artifacts (grainy feature in Figures 3b, 4b, 6b, 11b, 10b). This explains the lower value attained in SSIM and PSNR metrics in most of the images by HE compared to other methods. Our method consistently achieved near 1 SSIM value (worst is > 0.7) compared to the other three. This indicates our method will introduce fewer artifacts (grainy feature, emphasized boundary, whitened area, etc.).

We can realize that only one metric does not show all sides of improvement by the same methods. The visual quality of the image is a subjective manner. For a more complete and subjective assessment of the visual quality, we have conducted a survey on our method enhanced images which we present in the next section.

6.3 Survey

6.3.1 Survey Preparation. We have prepared an anonymous survey to get a quantitative opinion on our enhanced images. In the survey, we have placed 26 enhanced images (produced by our technique) side-by-side with their original image. The survey participant needs

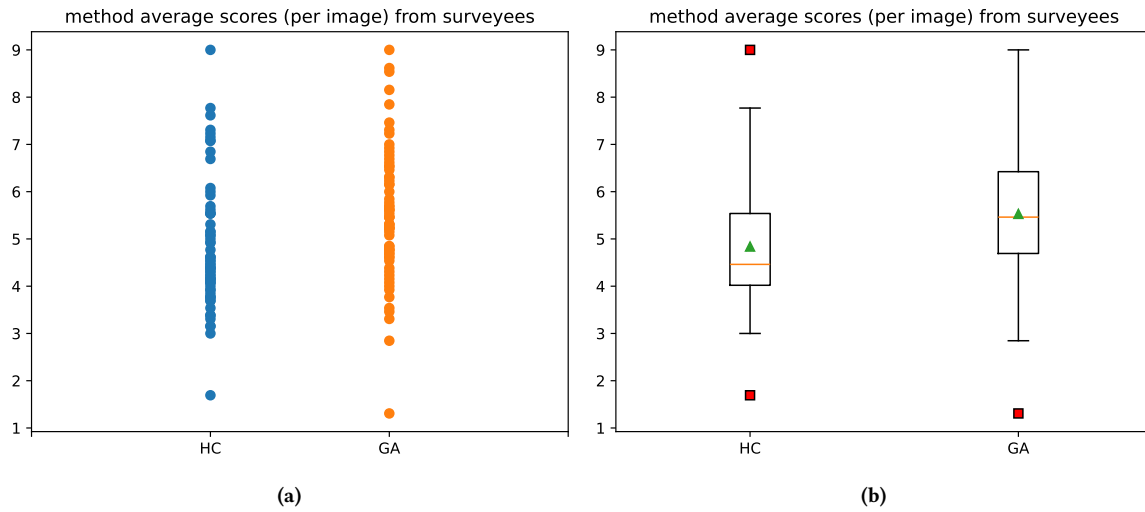


Figure 2: Average scores assigned to images enhanced by the HC and GA methods, as rated by survey participants

to mark the enhanced image on a scale of 1–9. As the reference, the mark of the original image was 5; therefore, a mark of > 5 would mean a better quality enhanced image, while < 5 will mean image degradation. A mark of 5 would mean the processed image is the same as the original visually (a subjective judgment).

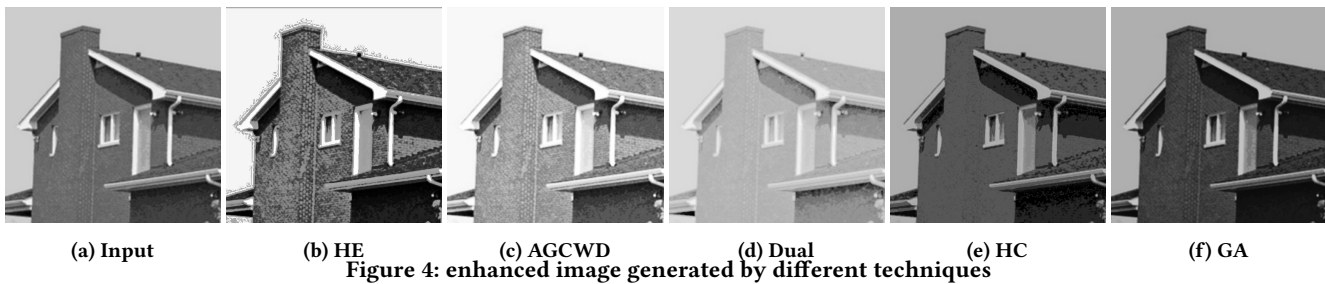
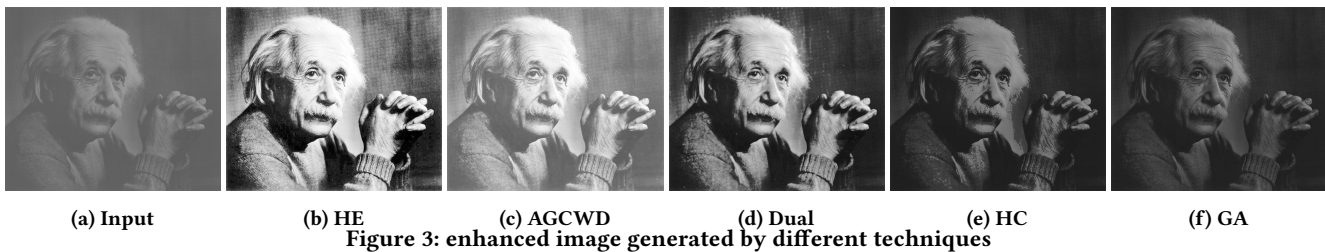
6.3.2 Survey Response & Observations. We have received a total of 67 responses. From the responses, we have filtered responses with suspicious patterns like all equal marks or very high or very low marks. This filtering caused two responses to be filtered. From the remaining 65 responses, it seems GA, achieving a mean score of 5.45, is better than HC, achieving a mean score of 4.77. Moreover, achieving a mean score of greater than 5 indicates that the GA improves the original image.

From Figure 2, we can see the mean scores of all enhanced images given by each surveyee. Each data point in Figure 2(a) indicates

the mean of scores given by one participant. Figure 2(b) shows the mean (triangle inside each box, green in color picture), and median (horizontal line inside each box, orange in color picture) of the responses. The red points are outliers according to the $1.5 \times \text{IQR}$ rule [38]. From Figure 2(a), GA shows a better mean score than HC (mean over participants). From Figure 2(b), we can see that HC has a higher chance of degrading the image (dense below score 5), although GA has achieved the image with the lowest score according to one participant.

7 Conclusion

In this work, we applied metaheuristics to develop an effective image-specific fuzzy logic-based transformation function. We utilized a quality function from previous studies to guide our enhancements. Recognizing that image quality is subjective, we conducted



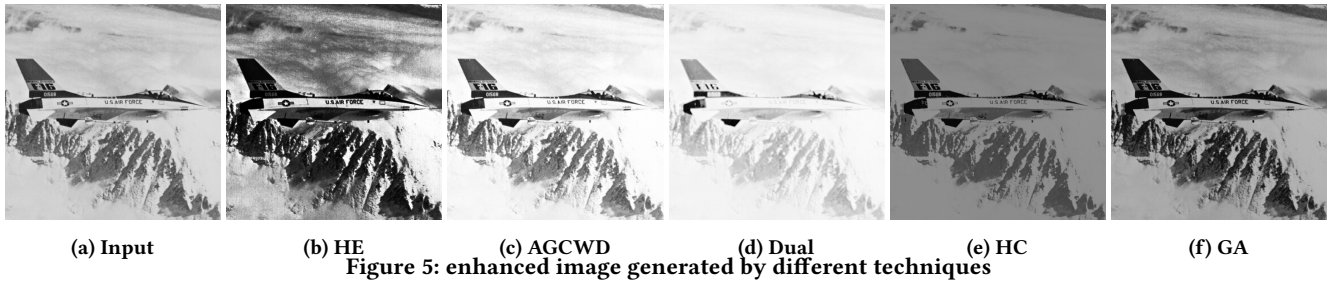


Figure 5: enhanced image generated by different techniques

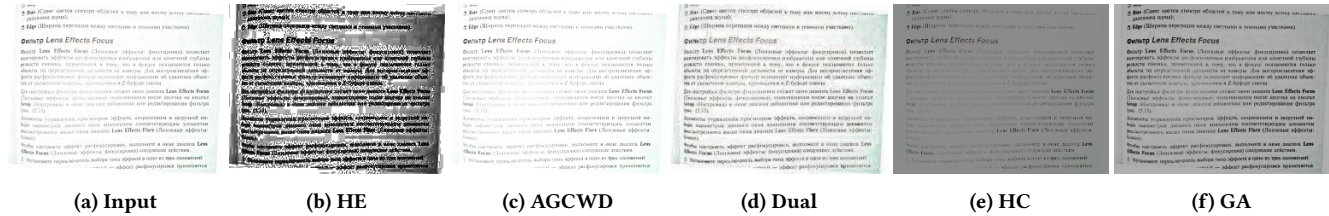


Figure 6: enhanced image generated by different techniques

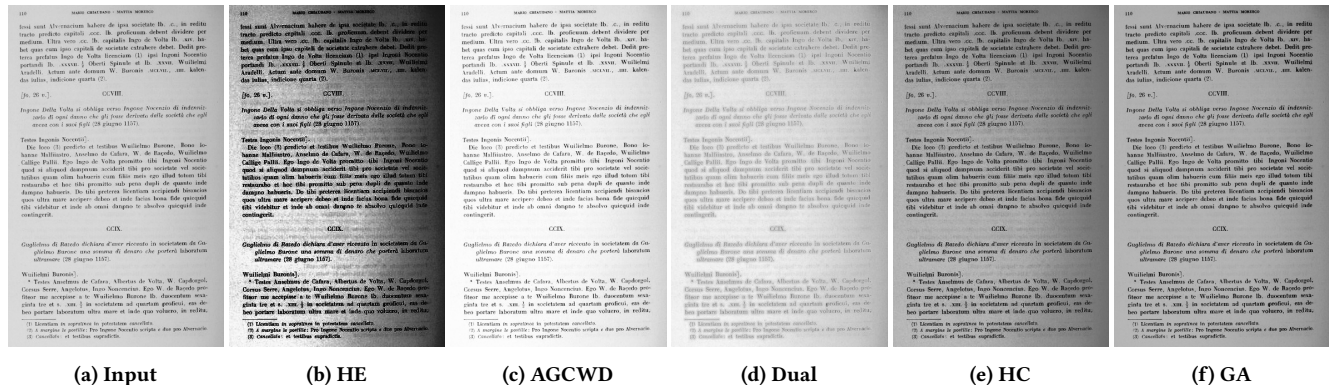


Figure 7: enhanced image generated by different techniques

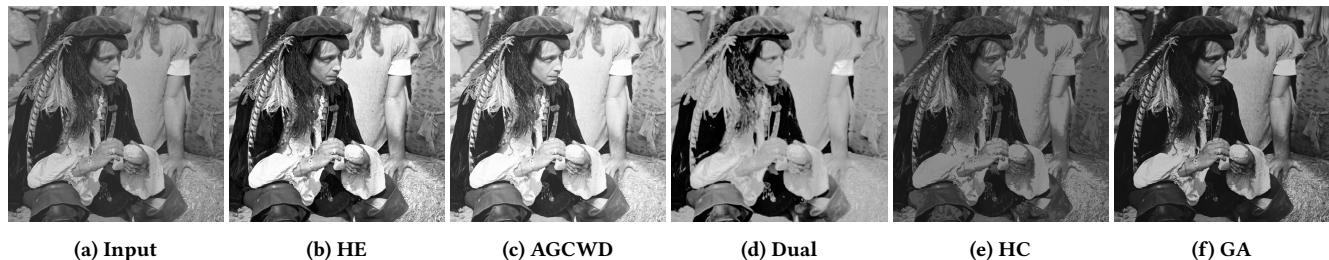


Figure 8: enhanced image generated by different techniques

a survey to gather subjective opinions on the visual quality of the enhancements and reported the results. The assessment revealed that one variant of our approach demonstrates visual improvement on average.

For future work, we plan to conduct additional experiments by incorporating more image processing operations, such as *gamma*

transformation, into the processing pipeline. One limitation of our current approach, based on visual observation, is that it tends to darken the image. Integrating other processing operations with automatic parameter tuning via metaheuristics holds promise for further improvement.

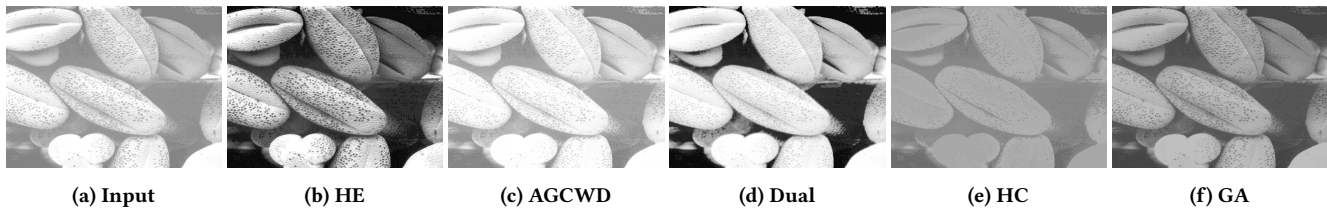


Figure 9: enhanced image generated by different techniques

Image	Contrast	Contrast					PSNR					SSIM				
		HE	AGC	Dual	HC	GA	HE	AGC	Dual	HC	GA	HE	AGC	Dual	HC	GA
Figure 3	0.307	1.000	0.482	1.000	0.711	0.738	12.178	14.059	14.200	13.115	13.191	0.46	0.76	0.62	0.65	0.74
Figure 4	1.000	1.000	1.000	1.000	1.000	0.913	18.272	15.081	12.480	18.316	16.313	0.67	0.86	0.81	0.78	0.86
Figure 5	0.894	1.000	0.889	1.000	1.000	0.909	12.077	17.231	12.559	12.967	15.260	0.55	0.96	0.79	0.84	0.9
Figure 6	0.802	1.000	0.802	1.000	0.893	0.903	6.322	31.501	22.375	7.553	9.689	0.29	0.95	0.89	0.61	0.89
Figure 7	1.000	1.000	1.000	1.000	1.000	0.912	9.554	18.066	21.031	13.239	15.925	0.52	0.98	0.84	0.88	0.95
Figure 8	0.774	1.000	0.752	1.000	0.791	0.843	18.220	14.389	17.006	17.681	18.644	0.87	0.87	0.72	0.76	0.83
Figure 9	0.339	1.000	1.000	1.000	0.262	0.500	10.598	22.720	13.632	14.787	12.978	0.67	0.93	0.77	0.69	0.91
Figure 10	0.360	1.000	0.360	0.727	0.320	0.399	7.442	41.648	33.793	6.510	11.110	0.3	0.99	0.98	0.7	0.94
Figure 11	0.917	1.000	1.000	1.000	0.908	0.903	8.438	27.879	16.463	6.850	13.004	0.48	0.97	0.63	0.69	0.94
Figure 12	1.000	1.000	1.000	1.000	0.923	0.913	6.845	33.446	22.716	13.545	13.161	0.24	0.98	0.88	0.93	0.95
Figure 13	0.925	1.000	0.931	1.000	0.902	0.909	9.614	17.897	15.826	12.781	14.873	0.53	0.98	0.79	0.91	0.95
Figure 14	1.000	1.000	0.992	1.000	0.889	0.903	24.806	16.212	10.564	13.742	17.269	0.87	0.64	0.89	0.66	0.82
Figure 15	0.836	1.000	0.821	1.000	0.916	0.886	19.460	14.725	12.843	16.009	16.740	0.88	0.88	0.76	0.79	0.84

Table 1: Michelson contrast, PSNR, and SSIM of proposed methods, histogram equalization (HE), dual channel (Dual), and adaptive gamma correction enhanced images (AGC), the contrast of reference image is in the second column. The image column contains the references to the input images.

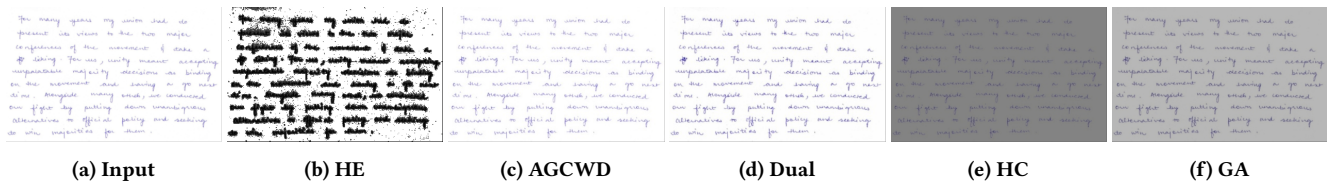


Figure 10: enhanced image generated by different techniques

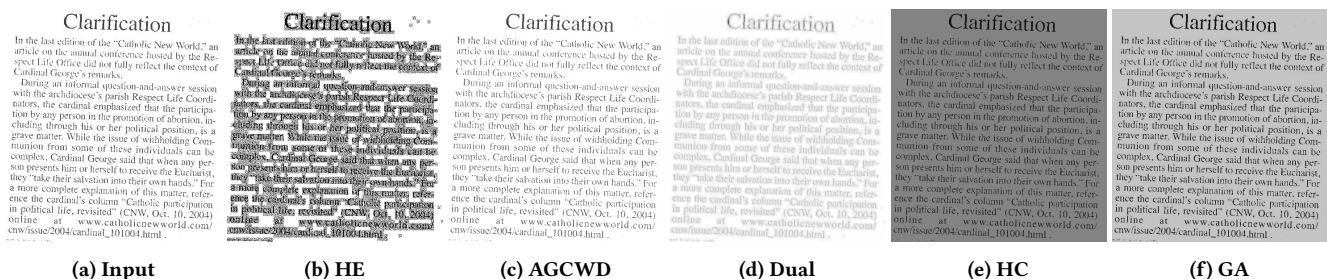


Figure 11: enhanced image generated by different techniques

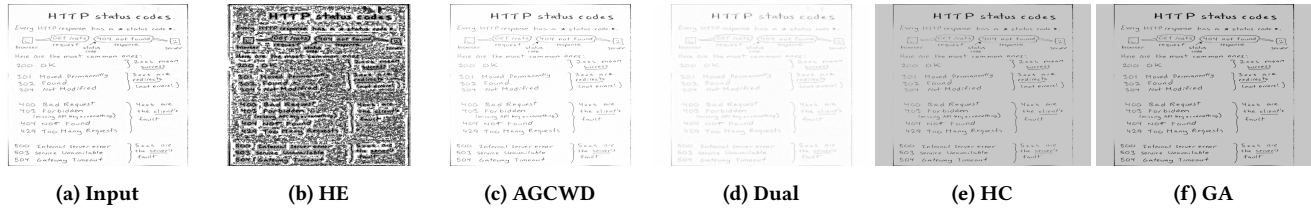


Figure 12: enhanced image generated by different techniques

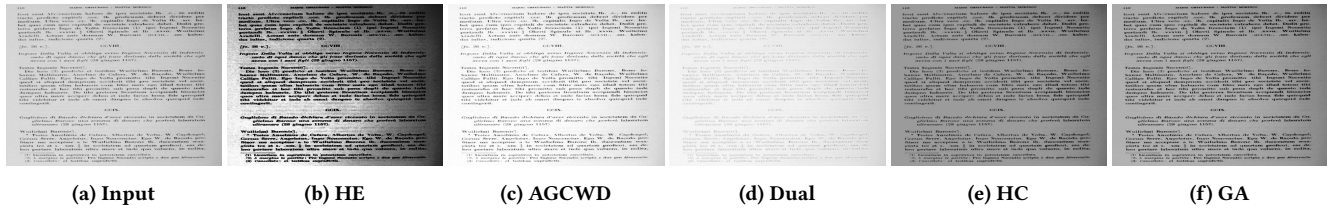


Figure 13: enhanced image generated by different techniques

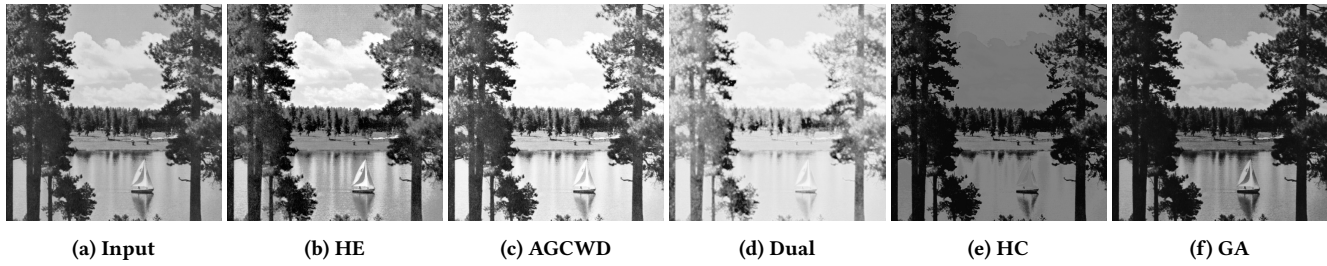


Figure 14: enhanced image generated by different techniques



Figure 15: enhanced image generated by different techniques

Acknowledgments

M Sohul Rahman is supported by a Basic Research Grant from BUET.

References

- [1] Tarik Arici, Salih Dikbas, and Yucel Altunbasak. 2009. A histogram modification framework and its application for image contrast enhancement. *IEEE Trans. on image processing* 18, 9 (2009), 1921–1935.
- [2] Dominic Asamoah, Emmanuel Ofori, Stephen Opoku, and Juliana Danso. 2018. Measuring the performance of image contrast enhancement technique. *International Journal of Computer Applications* 181, 22 (2018), 6–13.
- [3] Md Aashikur Rahman Azim, Mohimenul Kabir, and M Sohul Rahman. 2018. A simple, fast, filter-based algorithm for circular sequence comparison. In *International Workshop on Algorithms and Computation*. Springer, 183–194.
- [4] Kathrin Berkner, Michael J Gormish, and Edward L Schwartz. 2001. Multiscale sharpening and smoothing in Besov spaces with applications to image enhancement. *Applied and Computational Harmonic Analysis* 11, 1 (2001), 2–31.
- [5] Ashish Kumar Bhandari and Shubham Maurya. 2020. Cuckoo search algorithm-based brightness preserving histogram scheme for low-contrast image enhancement. *Soft Computing* 24 (2020), 1619–1645.
- [6] Ashish Kumar Bhandari, Syed Shah Nawazuddin, and Ayur Kumar Meena. 2019. A novel fuzzy clustering-based histogram model for image contrast enhancement. *IEEE Trans. on Fuzzy Systems* 28, 9 (2019), 2009–2021.
- [7] Malik Braik, Alaa Sheta, and Aladdin Ayeshe. 2007. Particle swarm optimisation enhancement approach for improving image quality. *International Journal of Innovative Computing and Applications* 1, 2 (2007), 138–145.
- [8] Margarita Arimatea Díaz-Cortés, Erik Cuevas, Jorge Gálvez, and Octavio Camarena. 2017. A new metaheuristic optimization methodology based on fuzzy logic. *Applied Soft Computing* 61 (2017), 549–569.
- [9] Leandro dos Santos Coelho, João Guilherme Sauer, and Marcelo Rudak. 2009. Differential evolution optimization combined with chaotic sequences for image

- contrast enhancement. *Chaos, solitons & fractals* 42, 1 (2009), 522–529.
- [10] Amer Draa and Amira Bouaziz. 2014. An artificial bee colony algorithm for image contrast enhancement. *Swarm and Evolutionary computation* 16 (2014), 69–84.
 - [11] Alberto Fernandez, Victoria Lopez, María José del Jesus, and Francisco Herrera. 2015. Revisiting evolutionary fuzzy systems: Taxonomy, applications, new trends and challenges. *Knowledge-Based Systems* 80 (2015), 109–121.
 - [12] Fred W Glover and Gary A Kochenberger. 2003. *Handbook of metaheuristics*. Vol. 57. Springer Science & Business Media.
 - [13] Rafael C Gonzalez. 2009. *Digital image processing*. Pearson education india.
 - [14] Chen Guojin, Zhu Miaofen, and Zhang Kesong. 2007. Study on definition evaluation function based on image contrast variation. (2007).
 - [15] Sara Hashemi, Soheila Kiani, Navid Noroozi, and Mohsen Ebrahimi Moghaddam. 2010. An image contrast enhancement method based on genetic algorithm. *Pattern Recognition Letters* 31, 13 (2010), 1816–1824.
 - [16] Horst Hauß Becker and Hamid R Tizhoosh. 2000. Fuzzy image processing. In *Computer vision and applications*. Elsevier, 541–576.
 - [17] Pourya Hoseini and Mahrokh G Shayesteh. 2013. Efficient contrast enhancement of images using hybrid ant colony optimisation, genetic algorithm, and simulated annealing. *Digital Signal Processing* 23, 3 (2013), 879–893.
 - [18] Shih-Chia Huang, Fan-Chieh Cheng, and Yi-Sheng Chiu. 2012. Efficient contrast enhancement using adaptive gamma correction with weighting distribution. *IEEE Trans. on Image Processing* 22, 3 (2012), 1032–1041.
 - [19] Andrey Ignatov, Radu Timofte, Thang Van Vu, Tung Minh Luu, Trung X Pham, Cao Van Nguyen, Yongwoo Kim, Jae-Seok Choi, Munchul Kim, Jie Huang, et al. 2018. Pirm challenge on perceptual image enhancement on smartphones: Report. In *ECCV Workshops*. 0–0.
 - [20] Kazi Ashik Islam, Ibraheem Muhammad Moosa, Jaiaid Mobin, Muhammad Ali Nayeem, and M Sohel Rahman. 2019. A heuristic aided Stochastic Beam Search algorithm for solving the transit network design problem. *Swarm and Evolutionary Computation* 46 (2019), 154–170.
 - [21] Yi Jin and Wei Wei. 2022. Image edge enhancement detection method of human-computer interaction interface based on machine vision technology. *Mobile Networks and Applications* 27, 2 (2022), 775–783.
 - [22] Sandeep Joshi and Samrudh Kumar. 2018. Image contrast enhancement using fuzzy logic. *arXiv preprint arXiv:1809.04529* (2018).
 - [23] Mohimenul Kabir, Flavio O Everardo, Ankit K Shukla, Markus Hecher, Johannes Klaus Fichte, and Kuldeep S Meel. 2022. ApproxASP—a scalable approximate answer set counter. In *AAAI*, Vol. 36. 5755–5764.
 - [24] Mohimenul Kabir and Kuldeep S Meel. 2024. On Lower Bounding Minimal Model Count. *arXiv preprint arXiv:2407.09744* (2024).
 - [25] Mohimenul Kabir, Jaiaid Mobin, Muhammad Ali Nayeem, Muhammad Ahsanul Habib, and M Sohel Rahman. 2023. Multi-objective optimization and heuristic based solutions for evacuation modeling. *Transportation research interdisciplinary perspectives* 18 (2023), 100798.
 - [26] Marcin Korytkowski, Roman Senkerik, Magdalena M Scherer, Rafal A Angryk, Mirosław Kordos, and Agnieszka Siwocha. 2020. Efficient image retrieval by fuzzy rules from boosting and metaheuristic. *Journal of Artificial Intelligence and Soft Computing Research* 10, 1 (2020), 57–69.
 - [27] MultiMedia LLC. 2024. *OpenCV: Changing Colorspaces*. https://docs.opencv.org/4.x/d9d/tutorial_py_colorspaces.html
 - [28] Martin Loesdau, Sébastien Chabrier, and Alban Gabillon. 2014. Hue and saturation in the RGB color space. In *ICISP*. Springer, 203–212.
 - [29] Michael Lones. 2011. Sean Luke: essentials of metaheuristics.
 - [30] Albert Abraham Michelson. 1995. *Studies in optics*. Courier Corporation.
 - [31] Souradeep Mukhopadhyay, Sabbir Hossain, Samir Malakar, Erik Cuevas, and Ram Sarkar. 2023. Image contrast improvement through a metaheuristic scheme. *Soft Computing* 27, 18 (2023), 13657–13676.
 - [32] Cristian Munteanu and Agostinho Rosa. 2004. Gray-scale image enhancement as an automatic process driven by evolution. *IEEE Trans. on systems, man, and cybernetics, part B (cybernetics)* 34, 2 (2004), 1292–1298.
 - [33] Adam Nevriyanto, Diah Purnamasari, et al. 2017. Image enhancement using the image sharpening, contrast enhancement, and Standard Median Filter (Noise Removal) with pixel-based and human visual system-based measurements. In *ICECOS*. IEEE, 114–119.
 - [34] Muhtahir Oloyede, Gerhard Hancke, Hermanus Myburgh, and Adeiza Onumanyi. 2019. A new evaluation function for face image enhancement in unconstrained environments using metaheuristic algorithms. *EURASIP Journal on Image and Video Processing* 2019 (2019), 1–18.
 - [35] Chen Hee Ooi, Nicholas Sia Pik Kong, and Haidi Ibrahim. 2009. Bi-histogram equalization with a plateau limit for digital image enhancement. *IEEE Trans. on consumer electronics* 55, 4 (2009), 2072–2080.
 - [36] William K Pratt. 2007. *Digital image processing: PIKS Scientific inside*. Vol. 4. Wiley Online Library.
 - [37] Alessandro Rizzi, Thomas Algeri, Giuseppe Medeghini, and Daniele Marini. 2004. A proposal for contrast measure in digital images. In *CGIV*, Vol. 2004. Society for Imaging Science and Technology, 187–192.
 - [38] Peter J Rousseeuw and Mia Hubert. 2011. Robust statistics for outlier detection. *Wiley interdisciplinary reviews: Data mining and knowledge discovery* 1, 1 (2011), 73–79.
 - [39] Sourav Samanta, Amartya Mukherjee, Amira S Ashour, Nilanjan Dey, João Manuel RS Tavares, Wahiba Ben Abdesslem Karâa, Redha Taïar, Ahmad Tahar Azar, and Aboul Ella Hassanien. 2018. Log transform based optimal image enhancement using firefly algorithm for autonomous mini unmanned aerial vehicle: An application of aerial photography. *International Journal of Image and Graphics* 18, 04 (2018), 1850019.
 - [40] Florian Schiller, Matteo Valsecchi, and Karl R Gegenfurtner. 2018. An evaluation of different measures of color saturation. *Vision research* 151 (2018), 117–134.
 - [41] Doron Shaked and Ingeborg Tastl. 2005. Sharpness measure: Towards automatic image enhancement. In *IEEE ICIP*, Vol. 1. IEEE, 1–937.
 - [42] P Shanmugavadivu and K Balasubramanian. 2014. Particle swarm optimized multi-objective histogram equalization for image enhancement. *Optics & laser technology* 57 (2014), 243–251.
 - [43] Zhenghao Shi, Mei Mei Zhu, Bin Guo, Minghua Zhao, and Changqing Zhang. 2018. Nighttime low illumination image enhancement with single image using bright/dark channel prior. *EURASIP Journal on Image and Video Processing* 2018 (2018), 1–15.
 - [44] Li Shuhua and Guo Gaizhi. 2010. The application of improved HSV color space model in image processing. In *ICFCC*, Vol. 2. IEEE, V2–10.
 - [45] Gursharn Singh, Anand Mittal, et al. 2014. Various image enhancement techniques—a critical review. *International Journal of Innovation and Scientific Research* 10, 2 (2014), 267–274.
 - [46] Ravindra Pal Singh and Manish Dixit. 2015. Histogram equalization: a strong technique for image enhancement. *International Journal of Signal Processing, Image Processing and Pattern Recognition* 8, 8 (2015), 345–352.
 - [47] J Alex Stark. 2000. Adaptive image contrast enhancement using generalizations of histogram equalization. *IEEE Trans. on Image Processing* 9, 5 (2000), 889–896.
 - [48] Shamik Sural, Gang Qian, and Sakti Pramanik. 2002. Segmentation and histogram generation using the HSV color space for image retrieval. In *ICIP*, Vol. 2. IEEE, II–II.
 - [49] Ahmad S Tarawneh, Ahmad B Hassanat, Issam Elkhadiri, Dmitry Chetverikov, and Khalid Almohammadi. 2020. Automatic gamma correction based on root-mean-square-error maximization. In *ICIT-1441*. IEEE, 1–5.
 - [50] A Vadivel, Shamik Sural, and Arun K Majumdar. 2005. Human color perception in the HSV space and its application in histogram generation for image retrieval. In *Color imaging X: processing, hardcopy, and applications*, Vol. 5667. SPIE, 598–609.
 - [51] Wencheng Wang, Xiaojin Wu, Xiaohui Yuan, and Zairui Gao. 2020. An experiment-based review of low-light image enhancement methods. *IEEE Access* 8 (2020), 87884–87917.
 - [52] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE Trans. on Image Processing* 13, 4 (2004), 600–612.
 - [53] Zhiwei Ye, Mingwei Wang, Zhengbing Hu, and Wei Liu. 2015. An adaptive image enhancement technique by combining cuckoo search and particle swarm optimization algorithm. *Computational intelligence and neuroscience* 2015, 1 (2015), 825398.
 - [54] Meng Yu, Liquan Shen, Zhengyong Wang, and Xia Hua. 2023. Task-Friendly Underwater Image Enhancement for Machine Vision Applications. *IEEE Trans. on Geoscience and Remote Sensing* (2023).
 - [55] Youlian Zhu and Cheng Huang. 2012. An adaptive histogram equalization algorithm on the image gray level mapping. *Physics Procedia* 25 (2012), 601–608.